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The reciprocal relation between lightning and pollution and their impact over Kolkata, India

Anirban Middey · Sutapa Chaudhuri

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Abstract Aerosol loading in the atmosphere can cause increased lightning flashes, and those lightning flashes produce NO_x , which reacts in sun light to produce surface ozone. The present study deals with the effect of surface pollutants on premonsoon (April–May) lightning activity over the station Kolkata (22.65° N, 88.45° E). Seven-year (2004–2010) premonsoon thunderstorms data are taken for the study. Different parameters like aerosol optical depth and cloud top temperature from the Moderate Resolution Imaging Spectroradiometer satellite products along with lightning flash data from Tropical Rainfall Measuring Mission's (TRMM) Lightning Imaging Sensor are analyzed. Some surface pollution parameters like suspended particulate matter, particulate matter 10, nitrogen oxides (NO_x), and surface ozone (O_3) data during the same period are taken account for clear understanding of their association with lightning activity. Heights of convective condensation level and lifting condensation level are collected from radiosonde observations to anticipate about cloud base. It is found that increased surface pollution in a near storm environment is related to increased lightning flash rate, which results in increased surface NO_x and consequently increased surface ozone concentration over the station Kolkata.

Keywords Lightning flash · Surface ozone · Pollution · Thunderstorm

Introduction

Pollution brings in both direct and indirect effect in weather and atmosphere. It is an undeniable fact that long-term aerosol pollution can affect cloud formation processes and rainfall (Li et al. 2011) and effectively modify weather systems (Trenberth et al. 2000). Various weather formation and their diverse development stages are affected by pollutant particles. How the variation in atmospheric pollutant affects the weather condition or how the degree of concentration varies with different stages of prevailing weather condition is still under consideration. Study of pollution meteorology over a particular station (Kolkata) for a given time span requires knowledge about the interconnectivity between various meteorological parameters and pollutant particles. Interrelation between lightning and pollution has become a much talk about issue over the globe. The effect of aerosols on cloud systems include direct adjustment of the solar radiation budget and indirect modification of the cloud dynamics and microphysical properties and have been discussed by a number of authors (Tie et al. 2003; Khain et al. 2005; Wang et al. 2011). Hobbs et al. (1974) found that pollutants that affect clouds are most likely to produce modifications in weather and climate. The consequence of aerosols on cloud microphysics is significant and comes about on a global scale (Bréon et al. 2002). Many studies relating to the effect of an urban area on the local weather activities have been done all over the globe (Changnon et al. 1981; Orville et al. 2001; Steiger et al. 2002; Soriano et al. 2001). Majority of these studies have attributed the effect to the urban heat island circulation with a possible role of air pollution. Pollution over the metropolis can elevate the cloud condensation nuclei (CCN) concentration, which in turn might produce changes in the microphysical processes taking place inside the clouds (Kar and Ha 2003). Among the various aspects of lightning research, cloud to ground

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A. Middey (✉) · S. Chaudhuri
Department of Atmospheric Sciences, University of Calcutta,
51/2 Hazra Road,
Kolkata 700 019, India
e-mail: anirban.middey@gmail.com

(CG) lightning has been one of the prime concerns of scientists. Westcott (1995) first recorded the evidences of enhanced CG lightning activity due to urban effect over the cities. Later on, Petersen and Rutledge (1998) found positive correlations between CG lightning flash density and rainfall amount. After that, many studies have been carried out exploring the effect of an urban area on the initiation and enhancements of CG lightning (Lyons et al. 1998; Orville et al. 2001; Soriano and Pablo 2002; Pinto et al. 2004). The changes in cloud microphysical processes will bring out changes in concentration, phase, and the size of cloud particles; as a consequence, the charge separation processes in thunderclouds are affected. According to Orville et al. (2001), the increased pollution in the boundary layer is anticipated to be functional in holding back the mean drop-let size, and more cloud water would therefore be active in separating the electric charge, causing more CG lightning flashes. The relationship between precipitation and lightning is found to be more consistent during premonsoon season compared to that of monsoon season over central Indian region (Lal and Pawar 2009). Qie et al. (2004) claim that there is a high ratio of lightning flash density to rain rate over the Tibet Plateau during premonsoon season than that during monsoon season, and Yuan and Qie (2008) established that mean cell-level flash rate during the premonsoon season is higher than that during monsoon season. Of late, Liou and Kar (2010) studied the relationship between precipitation and cloud-to-ground lightning over Taiwan, and they found that the values of rain yield per lightning flash are different for inland and coastal stations. Many studies in the recent years have shown that lightning activity in a region can be affected by changes in thermodynamical properties as well as increase in aerosol concentration (Khain et al. 2008; Bell et al. 2008, 2009).

The investigation is made in the present study to figure out the effect of atmospheric pollutants along with meteorological parameters on lightning activity and, conversely, association of lightning flash in producing surface ozone over Kolkata during premonsoon (April–May) thunderstorms (Chaudhuri 2008a, b, 2010a, b, 2011a, b). The study comprises of total lightning flash, not its individual components (cloud-to-ground, intra-cloud, and cloud-to-air) because the efforts are made in the present context to comprehend the impact of total lightning flashes in producing tropospheric NO_x and O_3 over a tropical metropolitan (Kolkata) in a qualitative way.

Materials and methods

The location of the study is Kolkata ($22^\circ 32' \text{N}$, $88^\circ 20' \text{E}$). The thunderstorm records are collected from the India Meteorological Department during the premonsoon

season (April–May) for the years 2004 to 2010. Heights of convective condensation level (CCL) and lifting condensation level (LCL) are computed with radiosonde/rawinsonde data collected from the www.weather.uwyo.edu website. Pollution data are collected from the West Bengal Pollution Control Board of India, which includes the concentration of suspended particulate matter (SPM), particulate matter 10 (PM_{10}), nitrogen oxides (NO_x), and ozone (O_3) over Kolkata. The lightning data are collected from the Lightning Imaging Sensor (LIS) database. The LIS is a Tropical Rainfall Measuring Mission's (TRMM) satellite-based instrument and detects cloud and ground lightning activities in the troposphere (Cecil et al. 2005). Total lightning flashes (intra-cloud and cloud-to-ground and cloud-to-air) are taken for the present study. The TRMM satellite has circled the Earth at an altitude of 350 km (the TRMM satellite was boosted from an altitude of 350–402 km in August 2001 to extend its mission life) with an inclination of 35° . The LIS sensor requires approximately 49 days returning to its original position relative to the Sun and the Earth, though the manual recommends using a 100-day window (Boccippio et al. 1998). LIS has a much smaller field of view but has higher detection efficiency than optical transient detector (Boccippio et al. 2002). Different parameters like aerosol optical depth (AOD) and cloud top temperature (CTT) are collected from the Moderate Resolution Imaging Spectroradiometer (aqua/terra) satellite products.

Results and discussion

The most significant weather phenomenon during the premonsoon season (April–May) over Kolkata ($22^\circ 32' \text{N}$, $88^\circ 20' \text{E}$) and its vicinity is the outburst of severe local storms (Chaudhuri and Middey 2011a) with intense lightning activity. CCL can be assumed as the theoretical cloud base, and its variation (Chaudhuri and Middey 2011b) represents

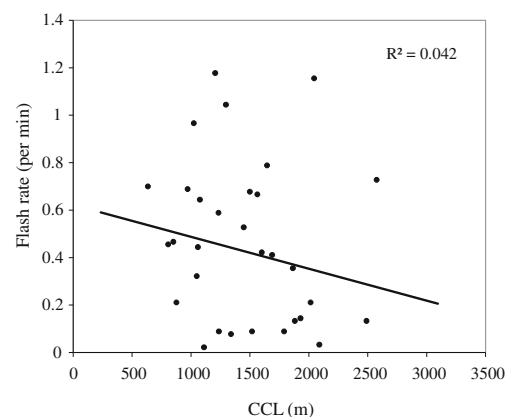
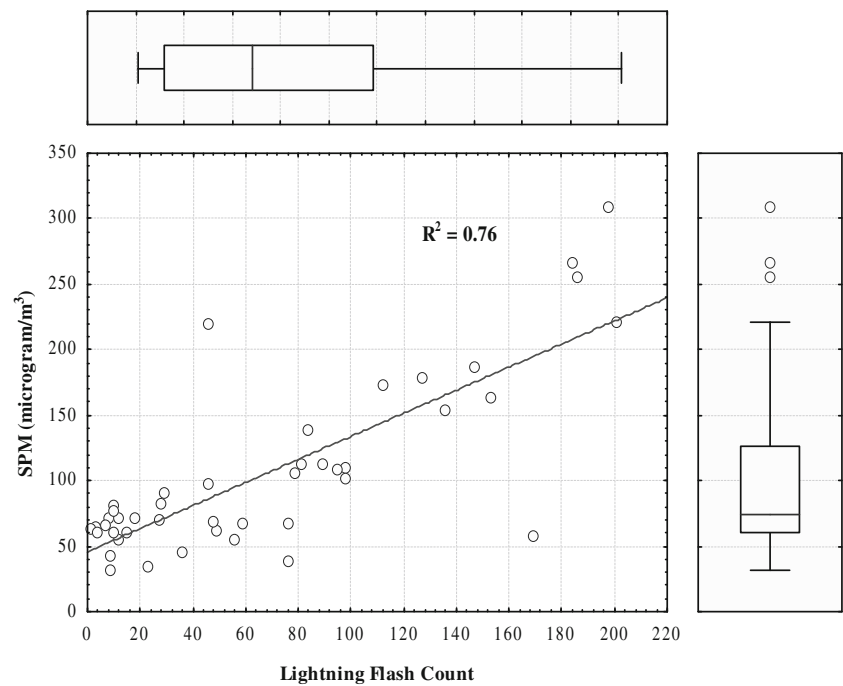


Fig. 1 Scatter plot of CCL vs. Flash rate during premonsoon thunderstorms over Kolkata

Fig. 2 Scatter plot of lightning flash count vs. SPM concentration along with their variability (*box-whisker*) during pre-monsoon thunderstorms over Kolkata



different cloud dynamics. The scatter plot of CCL against flash count shows very sparse representation (Fig. 1). Low value of CCL means that the cloud base formation is very near to the ground, and surface pollutants are easily available to get injected into the cloud with the help of strong wind flow. However, lightning flash count is not affected by high or low CCL values. On the contrary, if high concentration of SPM ($>240 \mu\text{g}/\text{m}^3$) prevails over the surface, lightning flash rate (per minute) are found to be high irrespective of CCL heights, as evident by good linear association between SPM and lightning flash rate (per minute; Fig. 2). Although SPM mainly concentrated at the values of $65\text{--}130 \mu\text{g}/\text{m}^3$, but whenever SPM values increase, the

lightning flash count also increases towards higher value (Fig. 2). The correlations between SPM and lightning flash rate can be elucidated by cloud microphysics. The higher SPM concentration over the surface contributes to more CCN formation, which results in the reduction of the mean cloud droplet size, by decreasing the process of coalescence and the droplet collision efficiency (Rogers and Yau 1989). Thus, more supercooled water is likely to exist at greater depths in clouds. The profusion of supercooled water may generate large graupel. The net effect is therefore an enhanced lightning activity over the urban regions (Liou and Kar 2010). Smaller particulate matter like PM_{10} (particles measuring $10 \mu\text{m}$ or less) also shows significant association with total number of lightning flashes over the years

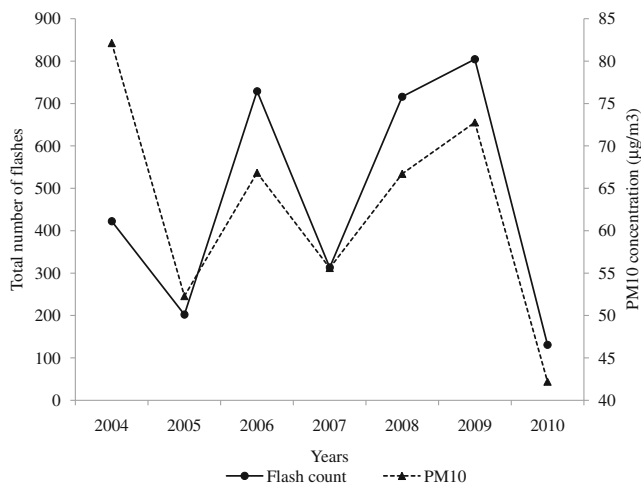


Fig. 3 Yearly variation of average particulate matter (PM_{10}) and total number of flashes over Kolkata during 2004–2010

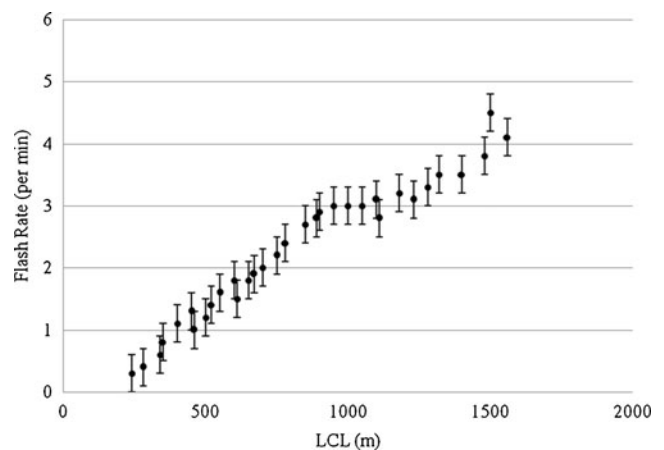


Fig. 4 Variation of lightning flash rate (per minute) with lifting condensation level (LCL) over Kolkata

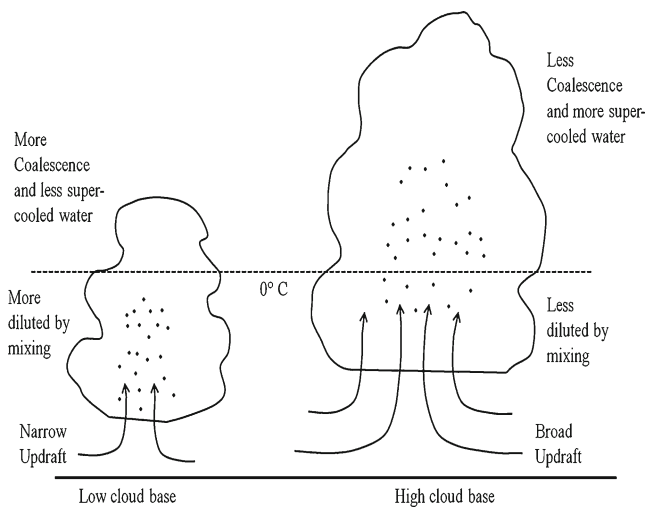


Fig. 5 Schematic showing the effect of cloud base on cloud microphysics (see William and Stanfill 2002)

(Fig. 3). During premonsoon period, Kolkata experiences many stationary thundercloud activities from local convective storms along with squall line travelling towards the stations, but they both cause high number of lightning flashes. The relation between cloud base height and lightning flash rate over Kolkata (Fig. 4) supports the observations of Williams and Stanfill (2002). Here, the LCL is considered as a theoretical cloud base (Craven et al. 2002), and the variation of lightning flash rate (per minute) with change in LCL is diagnosed. As the LCL value increases, lightning flash rate also increases over the station Kolkata. This relation between LCL and lightning flash rate is quite obvious because higher cloud base heights lead to larger updraft widths and reduced dilution by mixing (Williams et al. 2005), and more supercooled water is expected to exist at greater depths in clouds which in turn enhance the lightning activity (Fig. 5). NO_x concentrations are found to be at higher levels ($80 \mu\text{g}/\text{m}^3$) during premonsoon lightning activity over Kolkata (Fig. 6). Higher concentration of NO_x

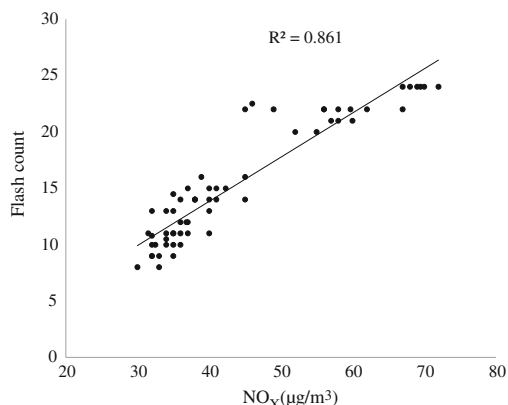


Fig. 6 Scatter plot of daily averaged flash count vs. NO_x during premonsoon months (April–May) over Kolkata

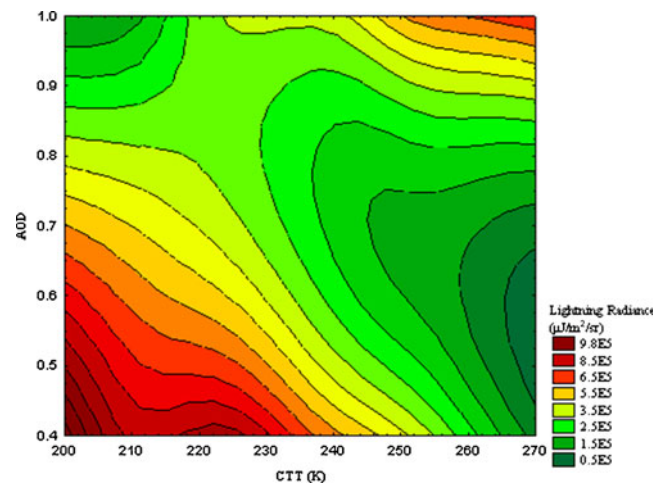


Fig. 7 Contour diagram showing the variation of lightning radiance with cloud top temperature and AOD values

(Zhou et al. 2005) may not be a predecessor for lightning rather than the combustion due to lightning flash stands responsible for NO_x ($\text{NO} + \text{NO}_2$) production in lower troposphere. As the lightning flash count increases, NO_x concentration also resides over higher values. The hydroscopic parts of near surface pollutants like SPM, PM_{10} , etc. play significant role in CCN formation. They move upward by virtue of winds and remains aloft as aerosols to take part in cloud formation. Concentration and various modes of aerosol (coarse or fine) determine cloud microphysics. Aerosol concentration plays a crucial role in producing lightning flash radiance. High value of lightning flash radiance can be explained by higher number of charge destroyed per unit area per unit time. This charge destruction process becomes faster in presence of opaque atmosphere (where aerosol optical depth is higher). Figure 7 represents a contour plot

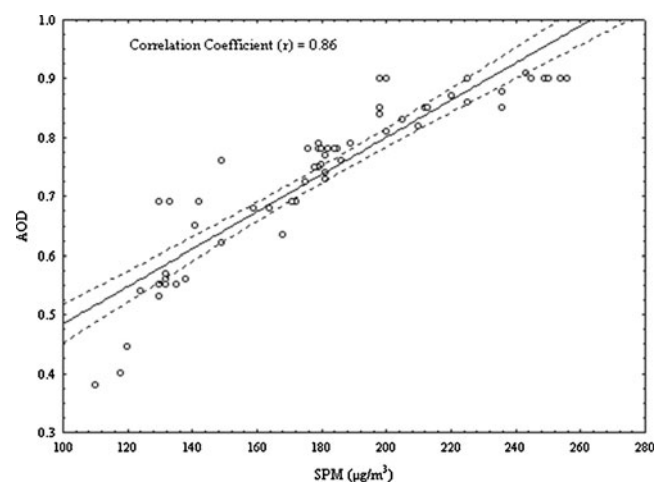


Fig. 8 Scatter diagram showing the correlation between ground-based observation of SPM and satellite based observation of AOD over Kolkata. Dashed lines represent 95 % confidence level bound

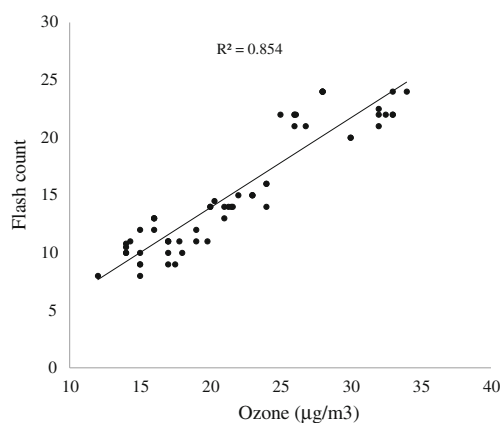


Fig. 9 Scatter plot of daily averaged flash count vs. surface Ozone during premonsoon months (April–May) over Kolkata

showing the variation of lightning flash radiance ($\mu\text{J m}^{-2} \text{sr}^{-1}$) with CTT and AOD. It is evident from the plot that at low CTT (which means cloud with greater height), lightning radiance is higher even at low AOD. But at high CTT values (represents smaller cloud height and relatively warm clouds), AOD must be high to obtain greater lightning radiance. According to Yuan et al. (2011), increase in aerosol loading results in reduced cloud ice particle sizes, and cloud glaciation is delayed to colder temperature, which in turn increases the lightning flashes by enhancing the cloud mixed phase activity. Lightning radiance is nothing but the energies from the charge destruction processes during lightning flash. As the aerosol loading increases in cloud mixed phase, the charge destruction process accelerates and consequently lightning radiance increases. It can be inferred from Fig. 7 that clouds with greater vertical extent (CTT ranges from 250 to 200 K) requires low to high AOD (0.4–0.8) for charge destruction processes to enhance lightning radiance. While very high

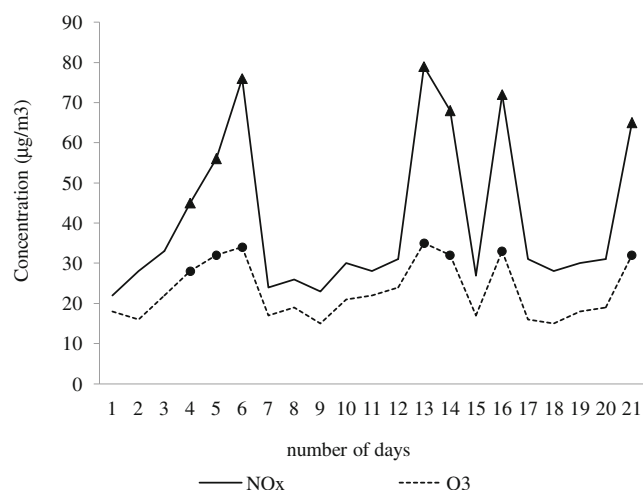


Fig. 10 Variation of NO_x and O_3 concentrations for few lightning activity day (marked) and no lightning days during premonsoon months over Kolkata

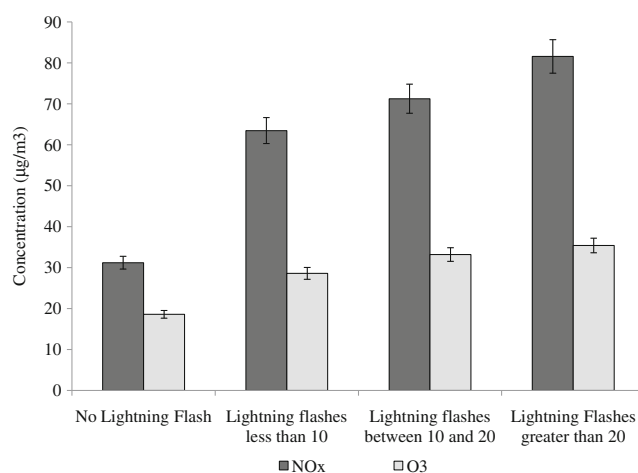


Fig. 11 Average concentrations of NO_x and O_3 for different ranges of daily lightning flashes along with no lightning flash during premonsoon seasons from 2004 to 2010

AOD (>0.85) is observed in clouds with small vertical extent (CTT <260 K) that plays a role in producing high lightning radiances. Greater amount of particulate matter from the surface is supposed to be injected in clouds with small vertical extent to produce more lightning radiances. Ground-based measurement of SPM shows very high correlation (correlation coefficient=0.86) with satellite-based AOD observation over Kolkata (Fig. 8) during thunderstorms of premonsoon seasons. In storm environment, huge amount of particulate matter from surface goes aloft with thunderstorm updraft and takes part in cloud microphysical processes (Rogers and Yau 1989). These particulate matters get infused in a thundercloud and serve as CCN; then, the cloud opacity is changed (as reflected by AOD values).

Thus, the higher concentration level of pollutant parameters near the surface can influence lightning activity and flash rate, and a higher number of lightning flashes stands

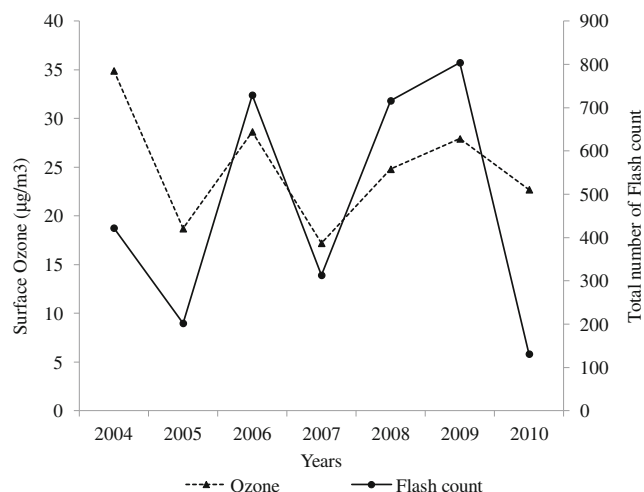


Fig. 12 Yearly variation of average surface ozone (O_3) and total number of flashes over Kolkata during 2004–2010

responsible for surface NO_x production over Kolkata. This nitrogen oxide gas (NO_x) reacts with sunlight and other gases in the atmosphere to produce ozone (O_3), which is harmful for human and plant health. The formation of tropospheric ozone in presence of NO_x can be summarized as follows (DeCaria et al. 2005);



A very good linear association is observed between lightning flash count and near surface ozone concentration over Kolkata (Fig. 9). However, lightning is not the only source of tropospheric NO_x . Other sources of NO_x include fossil fuel combustion, biomass burning, soil release, oxidation of atmospheric ammonia, and penetration of stratospheric NO_x . Over Kolkata, NO_x production seems to have good contribution from lightning flashes during premonsoon months. Few days with considerable amount of lightning flashes (>15) are taken along with some no lightning activity days for the analysis. It is found that the daily average values of NO_x and O_3 concentrations are much higher (Fig. 10) on the days with lightning flashes (>15) than on days with no lightning flash. Recently, Pawar et al. (2012) have found that lightning production of NO_x inside a thunderstorm can lead to significant impacts on surface ozone concentrations in the tropics. Lightning flashes are categorized in four ranges, viz., no lightning flash, lightning flash <10, lightning flashes between 10 and 20, and lightning flash >20, during thunderstorm events of 2004–2010 for the months of April and May. The daily averaged concentrations of NO_x and O_3 show markedly different values in those four cases (Fig. 11). NO_x concentration is found to be increased from no lightning day value of $31.2 \mu\text{g}/\text{m}^3$ to the value of $81.56 \mu\text{g}/\text{m}^3$ during the days with lightning flashes >20. Consequently, a concentration of O_3 is also observed higher ($35.4 \mu\text{g}/\text{m}^3$) for highest lightning activity days (lightning flashes >20) than that observed in no lightning days ($18.6 \mu\text{g}/\text{m}^3$). The yearly variation of surface O_3 concentrations from 2004 to 2010 are shown in Fig. 12 along with total number of lightning flashes during premonsoon months (April–May). It shows similar pattern for both the O_3 concentration and total flash count over the years.

Conclusion

The “NO_x-ious” effect (NASA, 2009) of lightning is also prominent over the station Kolkata. It is an irrefutable fact that pollution has a great connection with thunderstorm and lightning activity. Not only meteorological parameters but

also particulate matters in the troposphere play significant role in modification of such high frequency weather phenomena like thunderstorm and lightning activity. From this study, it is evident that SPM concentration at surface level can affect the lightning flash rate during premonsoon thunderstorms over Kolkata. Whether cloud base formed nearer to the surface or at higher level, SPM concentration >240 $\mu\text{g}/\text{m}^3$ contributes towards high amount of lightning flash rate. Near surface SPM and PM_{10} contribute significantly to increase AOD and hence serve as CCN and influence lightning activity. Enhanced lightning activity can increase the production of NO_x over Kolkata (or any other station). As evident from the present study, NO_x concentrations are found to be at higher levels ($80 \mu\text{g}/\text{m}^3$) during premonsoon thunderstorm and lightning activity. Lightning production of NO_x during thunder cloud activity can enhance surface ozone formation. Surface ozone, which is harmful pollutant, is found to be at higher concentration levels during significant lightning activity days, and it shares similar pattern with total number of lightning flashes over the years (2004–2010). Undoubtedly increased surface pollutants in near storm environment are related to increased lightning activity, which enhances NO_x concentrations in the near surface environment. This increased amount of NO_x , in the presence of sun light, takes part in photochemical reactions to produce surface O_3 . As a result, this reciprocal relationship of lightning and pollution is supposed to have significant impact over Kolkata.

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